

# STAT115 2024 Practice Exam Solutions

Compiled via Codex

**Q1: Answer: B.** Compute the sample mean:  $(36 + 49.1 + 41.5 + 40 + 46.4)/5 = 42.6$  kg.

**Q2: Answer: E.** Use the sample standard deviation with denominator  $n - 1$ :  $s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{4}} \approx 5.20$  kg.

**Q3: Answer: C.** The measurements are quantitative on a continuum, so the data type is continuous.

**Q4: Answer: A.** The 65 minutes is the observed sample mean from  $n = 764$  students, so it is a statistic and a realised value of a random variable.

**Q5: Answer: C.** The target population is all New Zealand high school students and the parameter is their mean weekly exercise time.

**Q6: Answer: B.** Treating 1989–2000 as 12 years, the incidence rate is  $\frac{175}{1904 \times 12} \times 100,000 \approx 7.66 \times 10^2$  fractures per 100,000 person-years, describing the study cohort.

**Q7: Answer: B.** The relative risk is  $(175/(1904 \times 12))/(54/(1373 \times 12)) \approx 2.34$ .

**Q8: Answer: D.** Under-diagnosis of men makes the observed male rate too small, so the ratio  $\text{rate}_{\text{women}}/\text{rate}_{\text{men}}$  is biased upward.

**Q9: Answer: A.** The quoted 4.5 and 1.9 values summarise the observed rate ratios in the study for the 60–69 and 80+ age bands.

**Q10: Answer: D.** The mean of the total is  $\mu_W = \mu_X + \mu_Y = 1750 + 750 = 2500$ .

**Q11: Answer: C.** With independent counts, variances add:  $\sigma_W = \sqrt{100^2 + 40^2} = \sqrt{11,600} \approx 1.08 \times 10^2$ .

**Q12: Answer: C.**  $\Pr(B) = 82/623 \approx 0.1316$ .

**Q13: Answer: A.**  $\Pr(A^c \cap B) = 50/623 \approx 0.0803$ .

**Q14: Answer: E.**  $\Pr(A^c \mid B) = 50/82 \approx 0.610$ .

**Q15: Answer: B.** Sensitivity is  $\Pr(T \mid B) = 0.97$ .

- Q16: Answer: A.** Specificity is  $\Pr(T^c \mid B^c) = 1 - 0.35 = 0.65$ .
- Q17: Answer: B.**  $\Pr(B \cap T) = \Pr(B) \Pr(T \mid B) = 0.20 \times 0.97 = 0.194$ .
- Q18: Answer: C.**  $\Pr(B^c \cap T) = 0.80 \times 0.35 = 0.28$ .
- Q19: Answer: A.**  $\Pr(T) = 0.194 + 0.28 = 0.474 \approx 0.47$ .
- Q20: Answer: C.** Positive predictive value  $= 0.194/0.474 \approx 0.41$ .
- Q21: Answer: B.** A binomial model requires a fixed  $n$ , identical success probability for each trial, independence, and two possible outcomes per trial.
- Q22: Answer: C.** From the table,  $\Pr(X = 3) = 0.0081$ .
- Q23: Answer: D.**  $\Pr(X > 3) = \Pr(4) + \Pr(5) = 0.00045 + 0.00001 = 0.00046$ .
- Q24: Answer: D.** Because 150 exceeds the mean of a symmetric normal distribution, the upper-tail probability must be between 0 and 0.5.
- Q25: Answer: A.** The upper tail is  $1 - \text{pnorm}(137.5, 127.5, 19.6)$ .
- Q26: Answer: B.** The middle probability is  $\text{pnorm}(134.5, 127.5, 19.6) - \text{pnorm}(119.5, 127.5, 19.6)$ .
- Q27: Answer: D.**  $z = (140.5 - 127.5)/19.6 \approx 0.66$ .
- Q28: Answer: B.** Using  $t_{0.975, 194}$  with  $s/\sqrt{n}$  constructs a 95% confidence interval for the population mean BMI.
- Q29: Answer: D.** Using  $z_{0.975}$   $s$  without  $\sqrt{n}$  gives an estimated 95% reference range for individual BMIs.
- Q30: Answer: C.** The mean paired difference (after – before) is  $52.9 - 56.5 = -3.6$  bpm.
- Q31: Answer: E.** The estimated standard error is  $s_d/\sqrt{n} = 5.9/\sqrt{13} \approx 1.64$  bpm.
- Q32: Answer: A.** For a 95% interval with  $n = 13$ , use  $t_{0.975, 12}$ .
- Q33: Answer: D.** The 95% CI for the population mean difference excludes zero and is entirely negative, so the data suggest a true decrease in resting heart rate among female runners following the protocol.
- Q34: Answer: A.** The sample proportion is  $\hat{p} = 49/99 \approx 0.495$ .
- Q35: Answer: B.**  $\text{SE} = \sqrt{\hat{p}(1 - \hat{p})/n} = \sqrt{0.495 \times 0.505/99} \approx 0.050$ .
- Q36: Answer: B.** A large-sample CI for a proportion uses  $\hat{p} \pm z_{0.975} \text{SE}$ .

- Q37: Answer: C.** The interval gives a 95% confidence range for the population proportion of New Zealand adults whose main news source is mainstream media.
- Q38: Answer: E.** A valid normal approximation requires a large sample and success probability not too close to 0 or 1;  $\hat{p} \approx 0.5$  satisfies this.
- Q39: Answer: B.** If mainstream-media users respond more, the sample proportion is inflated—a form of selection bias upward.
- Q40: Answer: C.** Two independent cross-sectional surveys describe prevalence; no exposure is manipulated, so this is descriptive.
- Q41: Answer: A.**  $\hat{p}_2 - \hat{p}_1 = 110/990 - 180/970 \approx -0.074$ .
- Q42: Answer: D.**  $SE = \sqrt{\hat{p}_1(1 - \hat{p}_1)/970 + \hat{p}_2(1 - \hat{p}_2)/990}$ .
- Q43: Answer: D.** A 99% two-sided CI uses  $z_{0.995}$ :  $(\hat{p}_2 - \hat{p}_1) \pm z_{0.995} SE$ .
- Q44: Answer: A.** The entire 99% CI is negative, supporting a reduction in the proportion of daily smokers between 2010 and 2020.
- Q45: Answer: C.** Participants were randomised to intervention or control, so this is an experimental analytic randomised controlled trial.
- Q46: Answer: C.** To compare population means:  $H_0: \mu_I = \mu_C$  versus  $H_A: \mu_I \neq \mu_C$ .
- Q47: Answer: B.**  $SE = \sqrt{24/102 + 12.5/91} \approx 0.61$  units of reduction.
- Q48: Answer: A.** Test statistic  $t = (9.6 - 8.3)/0.61 \approx 2.13$ .
- Q49: Answer: D.** For a two-sided  $t$ -test use  $2(1 - \text{pt}(|t|, \nu))$ .
- Q50: Answer: C.** With  $p = 0.0345 < 0.05$  we reject  $H_0$  and conclude there is evidence of a difference in mean mercury reduction between groups.
- Q51: Answer: D.** The  $p$ -value is the probability, under  $H_0$ , of observing a difference at least as extreme as 1.3 units (in absolute value).
- Q52: Answer: C.** Increasing both the sample sizes and the significance level (e.g.  $\alpha = 0.1$ ) increases statistical power.
- Q53: Answer: B.** Farmer odds  $= 53/447 \approx 0.119$ .
- Q54: Answer: D.** Non-farmer odds  $= 27/473 \approx 0.057$ .
- Q55: Answer: A.** Odds ratio  $= (53/447)/(27/473) = (53 \times 473)/(447 \times 27) \approx 2.08$ .

- Q56: Answer: E.**  $SE_{\log OR} = \sqrt{1/53 + 1/27 + 1/447 + 1/473} \approx 0.2455$ .
- Q57: Answer: B.** Exponentiating  $(0.2499, 1.2121)$  yields  $(e^{0.2499}, e^{1.2121}) \approx (1.28, 3.36)$ .
- Q58: Answer: A.** For an odds ratio, bounds entirely above/below 1 indicate increased/decreased odds; intervals spanning 1 imply insufficient evidence.
- Q59: Answer: A.** Missed diagnoses among farmers reduce observed cases, biasing the OR downward; this is information bias.
- Q60: Answer: C.** If farmers are more often male and males have lower MDD risk, sex confounding would pull the crude odds ratio toward 1, masking part of the association.
- Q61: Answer: D.** The adjusted OR of 1.37 has a 95% CI that includes 1, so after adjustment the data do not show a statistically significant association.
- Q62: Answer: E.**  $H_0$ : diagnosis and paternal age are independent;  $H_A$ : there is an association.
- Q63: Answer: A.** Expected count  $= (20 \times 528)/1612 \approx 6.55$  children.
- Q64: Answer: B.** Degrees of freedom  $(r - 1)(c - 1) = (2 - 1)(3 - 1) = 2$ .
- Q65: Answer: C.** Right-tail  $p$ -value is  $1 - \text{pchisq}(13.30, \nu)$ .
- Q66: Answer: A.** With  $p = 0.0014 < 0.05$  we reject  $H_0$  and infer an association between paternal age and autism diagnosis.
- Q67: Answer: C.** Retinol level is modelled as the response variable and age as the explanatory variable.
- Q68: Answer: A.** The fitted slope is about 3.07 ng/mL per year, as shown in the regression output.
- Q69: Answer: C.** Test  $H_0: \beta_1 = 0$  versus  $H_A: \beta_1 \neq 0$  for the age slope.
- Q70: Answer: C.** For simple regression with  $n = 100$ , the  $t$ -statistic uses  $\nu = n - 2 = 98$  degrees of freedom.
- Q71: Answer: E.** The  $p$ -value  $0.0172 < 0.05$  provides evidence of an association between age and retinol levels.
- Q72: Answer: E.** The intercept estimates the mean retinol level when age = 0, i.e., for a newborn.
- Q73: Answer: B.**  $\hat{\beta}_1$  estimates the mean change in retinol level for each additional year of age.
- Q74: Answer: C.**  $R^2 = 0.05654$  indicates that about 5.654% of the variance in retinol levels is explained by age.

- Q75: Answer: D.** Predicted value:  $441.86 + 3.07 \times 50 \approx 595.36$  ng/mL.
- Q76: Answer: B.** A 95% prediction interval gives the range where a new 40-year-old's retinol level would fall with probability 0.95.
- Q77: Answer: D.** Prediction errors grow as age moves away from the sample mean (50.61 years), so forecasts at 40.61 years are less precise.
- Q78: Answer: B.** Predicted for age = 42:  $441.86 + 3.07 \times 42 = 570.8$ ; residual =  $580.7 - 570.8 \approx 9.9$  ng/mL.
- Q79: Answer: A.** To model a binary outcome (cancer) with a continuous predictor, use logistic regression with cancer as the response.
- Q80: Answer: E.** The model predicts medical costs (charges), so charges is the response variable.
- Q81: Answer: B.**  $\hat{y} = -12052.46 + 257.73 \times 50 - 128.64 \times 1 + 322.36 \times 28 + 474.41 \times 3 + 23823.39 \times 0 \approx 1.12 \times 10^4$  dollars.
- Q82: Answer: B.**  $\hat{\beta}_{\text{bmi}}$  estimates the change in expected charges for each one-unit BMI increase while holding the other predictors fixed.
- Q83: Answer: A.** The CI for the age slope is  $257.73 \pm t_{0.975, 1332} \times 11.9$ .
- Q84: Answer: C.** The five predictors jointly explain about 74.97% of the variation in medical charges.
- Q85: Answer: C.**  $\mu_i$  denotes the population mean reduction for pesticide  $i$ , and  $e_{ij}$  is the residual for plot  $j$  using pesticide  $i$ .
- Q86: Answer: A.** ANOVA assumes  $e_{ij}$  are independent  $N(0, \sigma^2)$  within each group (common variance and zero mean).
- Q87: Answer: C.**  $H_0: \mu_A = \mu_B = \mu_C = \mu_D = \mu_E = \mu_F$  versus the alternative that not all population means are equal.
- Q88: Answer: B.** From the ANOVA table, total SS =  $2668.8 + 1015.2 = 3684.0$  and residual SS = 1015.2.
- Q89: Answer: D.** Large between-group variability relative to within-group noise favours rejecting  $H_0$  in one-way ANOVA.
- Q90: Answer: C.** The F-test  $p$ -value is `pf(f, 5, 66, lower.tail = FALSE)`.
- Q91: Answer: C.** The  $p$ -value  $< 0.01$  leads to rejecting  $H_0$ , indicating different mean reductions among at least some pesticides.

**Q92: Answer: B.** Blocking on country would absorb part of the residual variation, improving sensitivity to detect pesticide effects.